Digital Transmission & Line Coding Schemes

Coverage: Line Coding Criteria (DC, Sync, Baseline Wander), Unipolar, Polar (NRZ, Manchester), Bipolar (AMI), PCM & DM

Prepared by: Prof. Anju Markose | Assistant Professor, Dept. of Computer Science & Engineering | VJCET

1. Line Coding Schemes

- Manchester (IEEE 802.3): Mid-bit transition (1 = High-to-Low; 0 = Low-to-High). Zero DC and perfect clock sync.
- Differential Manchester: Mid-bit transition + start-bit transition for 0.
- Bipolar AMI: Bit 0 = 0V; Bit 1 = Alternating +V and -V.

2. Pulse Code Modulation (PCM)

- Sampling (fs >= 2*fmax) -> Quantization -> Encoding.

■ KTU University High-Yield Exam Questions

- Q: Draw line coding waveforms for bit stream 10110001 using NRZ-L, NRZ-I, Manchester, and AMI. (9 Marks)